Lab 02: Formatted I/O

CSE 4108

Structured Programming I Lab

Lab Tasks

1. **It's a Date:**

Write a program that accepts a date from the user in the format mm/dd/yyyy and then displays it in the form yyyymmdd.

Sample Run:

Enter a date (mm/dd/yyyy): 2/17/2021

You entered the data 20210217

2. **Inventory Management System:**

Mr. Sakamoto from Convenience Delivery Store (CDS) of Imperial University of Tokyo (IUT) is

back. This time, he wants you to create a list of items in his shop. He will provide you item

number, unit price and purchase date of the products available in CDS. He wants you to show

the items in an organized manner. However, before finalizing the design he wants to see how

the list will look like a single item.

You have to write a program that formats product information entered by the user. A session

with the program should look like this:

Sample Input:

Enter item number: 583

Enter unit price: 13.5

Enter purchase date (mm/dd/yyyy): 10/24/2019

Printing the receipt...

Item	Unit Price	Purchase Date
583	13.50	10/24/2019

The item number and date should be left justified; the unit price should be right justified. Allow dollar amounts up to $9999.99.

Hint: Use tabs (\t) to line up the columns

3. **Numbers Square:**

Write a program that asks the user to enter the numbers from 1 to 16 (in any order) and then

displays the numbers in a 4 by 4 arrangement, followed by the sums of the rows, columns,

and diagonals.

Sample Run:

Enter the numbers from 1 to 16 in any order:

16 3 2 13 5 10 11 8 9 6 7 12 4 15 14 1

Arranging as a square:

16 3 2 13

5 10 11 8

9 6 7 12

4 15 14 1

Row sums: 34 34 34 34

Column sums: 34 34 34 34

Diagonal sums: 34 34

Note: The underlined string is the input. Everything else is the printed output.

If the row, column, and diagonal sums are all the same (as they are in this example), the
numbers are said to form a magic square. The magic square shown here appears in a 1514
engraving by artist and mathematician Albrecht Dürer. (Note that the middle numbers in the
last row gives the date of the engraving).

4. **Know your ID:**

In Imperial University of Texas (IUT), a student ID is formatted as, +1569 AA B C DDD, where
+1569 is the University code, **AA** is the batch number, **B** is the department ID, **C** is the program ID and **DDD** is the student serial no.

Write a C program that would take a student ID as input in the format shown in the sample program and output the different components of the ID.

Sample Input:

Enter Student ID: +1569 16 4 1 048

Sample Output:

University Code: +1569

Batch Number: 16

Dept. ID: 4

Program ID: 1

Serial NO.: 048

5. **Fraction Addition:**

Modify the addfrac.c program (shown in class) so that the user enters both fractions at the same time, separated by a plus sign.

Sample Run:

```
Enter two fractions separated by a plus sign: 5/6+3/4
The sum is 38/24
```

Hint: Fractional sum can be calculated using the following equation:

$$\frac{a}{b} + \frac{c}{d} = \frac{(a \times d) + (b \times c)}{b \times d}$$